

TASK 3 — ANALYSIS OF VIRTUAL TRY-ON DATASETS

Name: Praneetha AS (aspraneetha29@gmail.com)

Role: AI/ML Intern

Date: 15/05/2026

Project Folder: Raritone

1. Purpose of Dataset Research

The objective of this task was to study datasets used in Artificial Intelligence virtual try-on systems and understand how training data improves AI-based clothing simulation models.

The research focused on:

- fashion image datasets
- clothing segmentation datasets
- body landmark datasets
- pose estimation datasets
- AI training workflows

The task also explored how datasets are used to train machine learning models for virtual clothing applications.

2. Introduction to Virtual Try-On Datasets

Virtual try-on datasets are collections of images and annotations used to train Artificial Intelligence systems for digital clothing simulation.

These datasets contain:

- human body images
- clothing images
- pose landmarks
- segmentation masks
- body measurements
- fashion categories

AI systems use these datasets to learn:

- clothing alignment

- body tracking
- cloth simulation
- human pose estimation
- garment fitting



Figure 1 — Virtual Try-On Dataset Samples



3. Components of AI Fashion Datasets

A virtual try-on dataset generally includes multiple components.

Human Body Images

Images of people in different poses and clothing styles are collected.

The dataset contains:

- front view poses
 - side poses
 - standing positions
 - walking positions
-

Clothing Images

Separate clothing images are stored for AI garment simulation.

These include:

- shirts
 - pants
 - jackets
 - dresses
 - footwear
-

Pose Landmark Data

Body landmark coordinates are stored for pose estimation systems.

Landmarks include:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

Segmentation Masks

Segmentation masks separate:

- body regions
- clothing regions
- background areas

This improves clothing alignment accuracy.

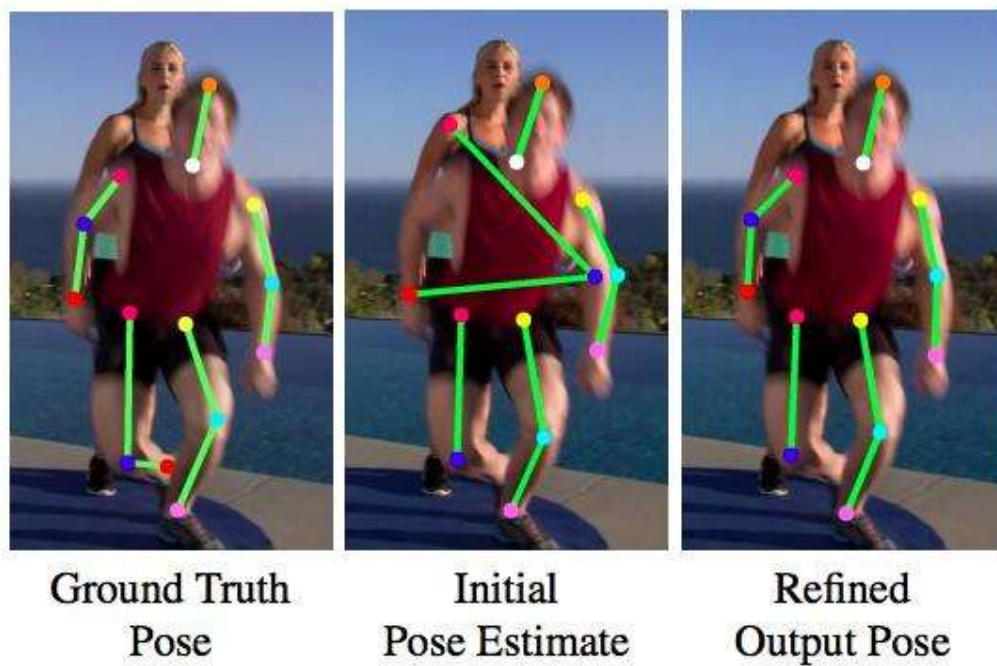


Figure 2 — Human Pose Dataset Annotation

4. Popular Virtual Try-On Datasets

Dataset Name	Purpose
DeepFashion	Fashion image dataset
VITON Dataset	Virtual try-on training
Fashion-MNIST	Clothing classification
ATR Dataset	Human parsing
CIHP Dataset	Human segmentation

These datasets help AI systems improve:

- clothing fitting
 - segmentation
 - body tracking
 - fashion recommendation
-

5. AI Training Workflow Using Datasets

The AI model training workflow generally includes:

Step 1 — Data Collection

Fashion images and body pose images are collected.

Step 2 — Annotation

Landmarks and clothing regions are labelled.

Step 3 — Dataset Preprocessing

Images are resized and normalized for AI training.

Step 4 — AI Model Training

Machine learning models learn body and clothing patterns.

Step 5 — Testing and Validation

The trained model is tested for fitting accuracy and prediction quality.



Figure 2. Example images of different categories and attributes in DeepFashion. The attributes form five groups: texture, fabric, shape, part, and style.

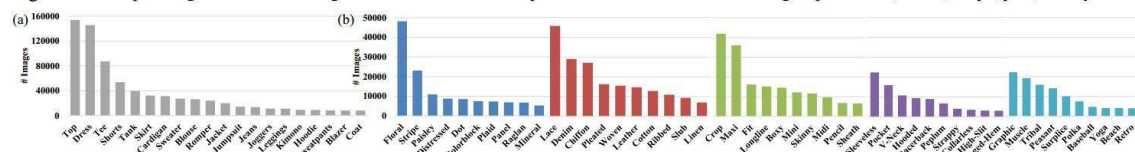


Figure 3. (a) Image number of the top-20 categories. (b) Image number of the top-10 attributes in each group.

Figure 3 — AI Dataset Training

6. Applications of Virtual Try-On Datasets

These datasets are used in:

- AI fashion systems
- virtual fitting rooms
- clothing recommendation platforms
- body segmentation systems
- augmented reality applications
- smart shopping systems

7. Benefits of AI Fashion Datasets

The advantages include:

- improved AI prediction accuracy
- better clothing alignment
- realistic garment simulation
- enhanced body tracking

- efficient AI model training
- improved virtual shopping systems

8. Challenges in Dataset Preparation

Challenge	Description
Large Dataset Size	Requires high storage
Annotation Complexity	Time-consuming labelling
Lighting Variations	Affects AI training
Pose Diversity	Requires multiple body positions
Dataset Imbalance	Reduces prediction quality

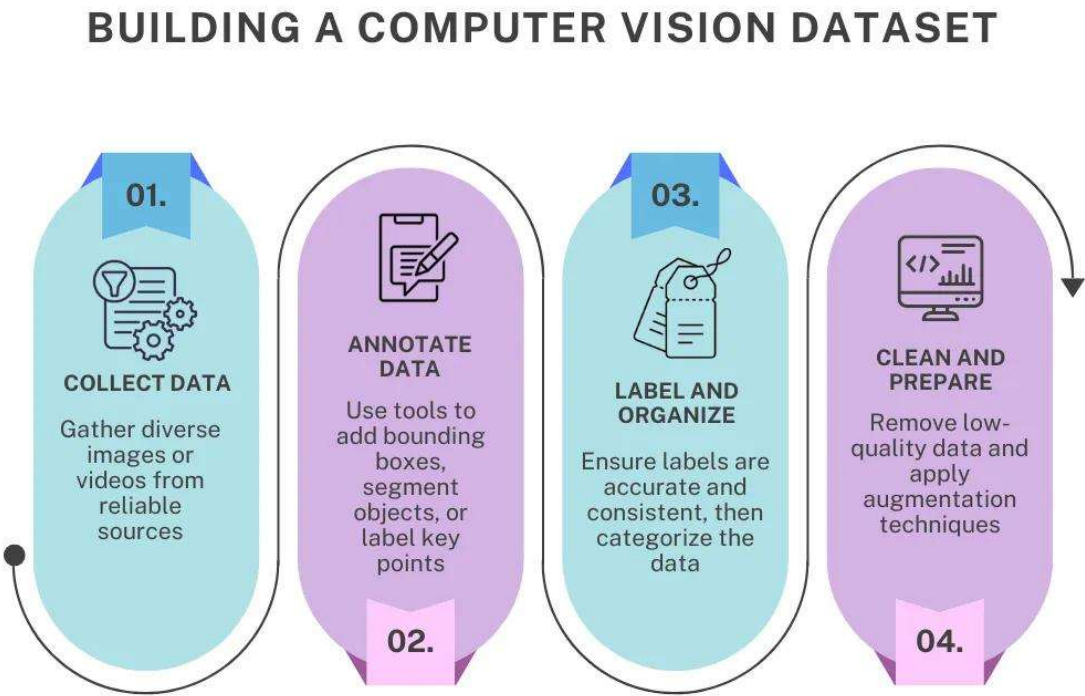


Figure 4 — Fashion Dataset Processing

9. Concepts Learned

During this research task, the following concepts were understood:

- AI dataset workflows
- fashion dataset annotation
- pose estimation datasets
- clothing segmentation systems
- machine learning training workflows
- virtual try-on systems

10. Final Summary

Virtual try-on datasets are essential for training Artificial Intelligence systems used in digital fashion applications.

This research improved understanding of:

- AI dataset preparation
- fashion image annotation
- clothing segmentation
- body landmark datasets
- machine learning workflows
- AI fashion technologies

These datasets play an important role in improving virtual fitting accuracy and intelligent shopping systems.